Spider Webs

01 Which lines are sticky on an orb web?



Strong dry spokes. A spiral that may hold glue.

Look closer. Find the center hub, the straight lines that go out like spokes, and the spiral that winds around them.

A little explanation. On many orb webs, the spokes (radii) and the small hub are dry and strong. The outer spiral often holds tiny glue beads. Spiders walk the dry lines and feel prey on the sticky spiral. Not every web is an orb, and not every spiral is sticky the same way.

Try this. From a few steps away, sketch a web you can see. Mark the hub, three spokes, and one loop of the spiral. Do not touch the silk.

02 How can silk catch without glue?



Glue beads, or woolly silk that snags.

Look closer. Find the two silk samples: one with shiny droplets, and one that looks soft and woolly.

A little explanation. Some spiders make capture silk with glue droplets (ecribellate silk). Others comb out a dry, woolly silk with a special plate called a cribellum (cribellate silk). The woolly silk snags legs and wings without sticky glue. Scientists still study exactly how each kind works best.

Try this. On your sketch, label one spiral loop 'maybe glue' and write one question you still have about silk.

03 How does a spider feel a tiny tug?



Vibrations travel along the silk.

Look closer. Find the spider resting near the hub, with one or two legs on the silk.

A little explanation. Silk carries vibrations. Many spiders keep legs on the radii so a buzz or tug can travel to them. Tiny sensors in the legs help them tell a breeze from a struggling insect. They usually do not need sticky feet on every line because they know the dry paths.

Try this. Tap a table lightly with a pencil, then harder. Tell which tap felt bigger through your hand. That is only a toy model of vibration, not a real web.

04 Why does dew make a web easy to see?



Dew catches light so we can see the pattern.

Look closer. Find the morning web with water beads shining on the spiral.

A little explanation. Cool mornings leave tiny water drops on silk. The drops catch light, so the web that was nearly invisible overnight suddenly shows its pattern. As the air warms, many drops shrink or fall. Dew helps us look; it is not glue from the spider.

Try this. On a cool morning, look for a web on a fence or bush. Count how many spiral loops you can see before the dew fades. Stay back and do not touch.